Assignment – 1.21

Review and Practice Practical

Q - 1. Open Cisco Packet Tracer and create a new project file named 'HomeWiFiNetwork.pkt' with three PCs and one wireless router connected appropriately.

- **Create a Wi-Fi Network:**
  1. Open **Cisco Packet Tracer**.
  2. Click **File → New**.
  3. Add **3 PCs** from End Devices.
  4. Add **1 Wireless Router** from Network Devices.
  5. Connect the PCs to the router using Wi-Fi or Ethernet cables.
  6. Click **File → Save As**.
  7. Save the file as **HomeWiFiNetwork.pkt**.
- **Result:** Your network will have 3 PCs connected to 1 wireless router.

Q - 2. Assign static IP addresses to each PC in your Packet Tracer network so they are on the same subnet (e.g., 192.168.10.x), and configure the default gateway to the router's IP.

➢ **Assign Static IP Addresses in Cisco Packet Tracer:**

1. Click **PC0**.
2. Go to **Desktop → IP Configuration**.
3. Select **Static** and enter:

- IP Address: 192.168.10.2
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1

4. Repeat the same steps for the other PCs using these settings:

Device	IP Address	Subnet Mask	Default Gateway
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.10.3	255.255.255.0	192.168.10.1
PC2	192.168.10.4	255.255.255.0	192.168.10.1

5. Click the **Wireless Router** and set its LAN IP address to 192.168.10.1 if it is not already configured.

Q - 3. Test connectivity by using the Packet Tracer 'ping' tool to verify that all PCs can communicate with each other and the router. Hint: Use the Command Prompt tool on each PC and type 'ping [other PC's IP]'.

- **Test Connectivity in Cisco Packet Tracer:**
- **Steps:**

1. Click on **PC0**.
2. Select **Desktop → Command Prompt**.
3. Type the following command and press Enter:

   ping 192.168.10.2

- **Test the other PCs:**
  - PC0: ping 192.168.10.2
  - PC0: ping 192.168.10.3
  - PC1: ping 192.168.10.3
  - PC0: ping 192.168.10.254 (Router)
- **Result:**

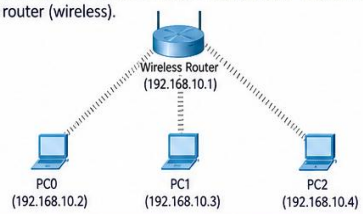
- **Successful:** Reply from 192.168.10.2 means the connection is working.
- **Failed:** Request timed out means you should check the IP addresses and connections.

Q - 4. Document each step you took to build and configure your Packet Tracer network, including screenshots of your topology and the IP configuration windows for each PC, in a Word or PDF file.

Home WiFi Network – Packet Tracer Configuration

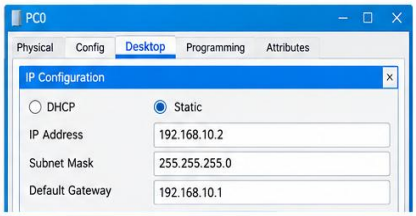
Step 1: Create the network

I added 3 PCs and 1 wireless router in Cisco Packet Tracer and connected the PCs to the router (wireless).



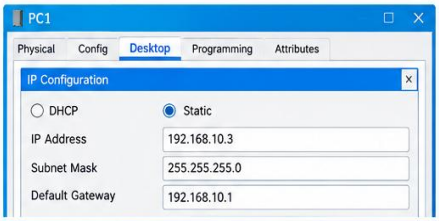
Step 2: Configure PC0 (IP Configuration)

I set the IP address, subnet mask and default gateway for PC0.



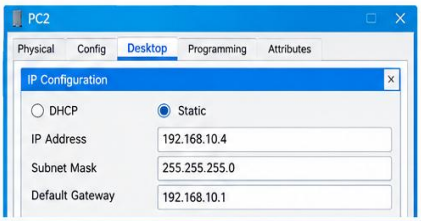
Step 3: Configure PC1 (IP Configuration)

I set the IP address, subnet mask and default gateway for PC1.



Step 4: Configure PC2 (IP Configuration)

I set the IP address, subnet mask and default gateway for PC2.



Step 5: Test Connectivity (Ping)

I opened the Command Prompt on each PC and pinged the other PCs and the router.

PC0 – Ping PC1

```
C:\>ping 192.168.10.3

Reply from 192.168.10.3: bytes=32
Reply from 192.168.10.3: bytes=32
Reply from 192.168.10.3: bytes=32
Reply from 192.168.10.3: bytes=32

Ping statistics for 192.168.10.3:
    Packets: Sent = 4,  Received = 4,
    Lost = 0 (0% loss).
```

PC1 – Ping PC2

```
C:\>ping 192.168.10.4

Reply from 192.168.10.4: bytes=32
Reply from 192.168.10.4: bytes=32
Reply from 192.168.10.4: bytes=32
Reply from 192.168.10.4: bytes=32

Ping statistics for 192.168.10.4:
    Packets: Sent = 4, Received = 4,
    Lost = 0 (0% loss).
```

PC2 – Ping Router

```
C:\>ping 192.168.10.1

Reply from 192.168.10.1: bytes=32
Reply from 192.168.10.1: bytes=32
Reply from 192.168.10.1: bytes=32
Reply from 192.168.10.1: bytes=32

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4,
    Lost = 0 (0% loss).
```

Step 6: Save the Project

1. Save the Packet Tracer file as **HomeWiFiNetwork.pkt**.
2. Take screenshots of the topology, IP configuration of all PCs and the ping results.
3. Include all screenshots in your Word or PDF report.

IP Address Summary

Router (LAN): 1c 2.168.10.1 / 255.255.255.0
PC0: 192.168.10.2 / 255.255.255.0 (Gateway: 192.168.10.1)
PC1: 192.168.10.3 / 255.255.255.0 (Gateway: 192.168.10.1)
PC2: 192.168.10.4 / 255.255.255.0 (Gateway: 192.168.10.1)